

ASHOK KUMAR L

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CAREER SUMMARY

Machine Learning Engineer with 2+ years of experience building production-grade AI automation systems — including presales pipeline automation, autonomous agentic workflows, advanced RAG architectures, and LLM fine-tuning. Skilled across the full AI product lifecycle: system design, event-driven backend development, document intelligence, LLM orchestration, and containerised deployment. Experienced with Apache Kafka, OpenAI APIs, SMTP automation, Docker Compose, and cloud-native services (OCI/Azure). Passionate about shipping reliable AI solutions that eliminate manual effort and deliver measurable business impact.

TECHNICAL SKILLS

AI / ML & LLM: Advanced RAG, Agentic Workflows, LangChain, LangGraph, LoRA/QLoRA Fine-tuning, OpenAI Vector Store

Document Intelligence: openpyxl, pdfplumber, reportlab, python-docx, python-pptx, lxml — Excel/PDF/Word/PPT/CSV processing

Event-Driven Systems: Apache Kafka (aiokafka), NiFi ConsumeIMAP → PublishKafka, SMTP automation, Tenacity retry

Programming & Frameworks: Python (FastAPI, asyncio, SQLAlchemy), SQL, Scikit-learn, TensorFlow, PyTorch

Cloud & Infrastructure: OCI, Azure Blob Storage, Docker Compose, Google Drive API, CI/CD Pipelines

Databases & Search: MySQL, Chroma DB, Elasticsearch, MariaDB, PostgreSQL

PROFESSIONAL EXPERIENCE

AI Chatbot Developer | Inswit Software Private Limited | Mar 2026 – Present

- **Voice Pipeline – Real-Time Voice Agent Platform:** Architected a multi-tenant, real-time voice agent backend on LiveKit Agents, with per-tenant STT/LLM/TTS provider selection and system prompts/tools/VAD settings loaded from MongoDB at call time — zero per-tenant code. Built a FastAPI WebSocket bridge integrating external Java IVR clients over raw PCM audio via LiveKit/WebRTC, a dynamic HTTP tool-calling layer for LLM function execution, and post-call analysis combining ai-coustics/Tyto acoustic scoring with automated call-quality metrics (tone, hesitation, confidence, pacing).
- **Presales Automation Pipeline:** Kafka consumer ingests inbound RFP emails, extracts requirements from multi-format attachments (Excel, Word, PDF, PPT, CSV) using AI, generates vendor responses via OpenAI vector store RAG, and writes answers back into the source document with colour-coded compliance labels (FC / PC / NC) — all without human input. Reduced presales response turnaround from ~3 business days to under 5 minutes; deployed on Docker Compose with FastAPI, SQLAlchemy/MySQL, and SMTP auto-reply.
- **LoRA Fine-tuning Pipeline:** Built a PEFT pipeline using LoRA/QLoRA with 4-bit quantization, boosting model accuracy from 59% to 89% (+30%) on domain-specific tasks while reducing GPU memory usage by ~60%. Covered end-to-end workflow: dataset prep, tokenization, SFT, and evaluation using HuggingFace Transformers and PEFT.
- **WebPage2PDF – Website Scraper & Document Generator:** Built a FastAPI service that crawls websites (BFS, depth 1–3, up to 50 pages) with Selenium JS-rendering fallback and BeautifulSoup4 semantic content extraction (headings, tables, images). Generates professionally formatted PDF (Playwright/Chromium) and DOCX (python-docx) with branded cover pages, auto-generated Table of Contents, and smart image sizing. Supports batch processing of up to 20 URLs with ZIP output; containerised with Docker Compose and a Streamlit UI frontend.

Machine Learning Engineer | Giggso India Pvt Ltd | Sep 2024 – Feb 2026

- **Autonomous Agentic Workflows:** Designed and deployed an agentic AI system using LangGraph/LangChain, reducing manual task intervention by 35%.
- **LLM-powered Email Automation:** Built LLM-powered email orchestration system, improving departmental response times by 50%.
- **Multimodal RAG Systems:** Engineered "NLWeb", a multimodal RAG pipeline with Chroma DB/Gemini, achieving 90%+ retrieval accuracy across YouTube transcripts and image data.
- **Production AI & MLOps:** Managed AI lifecycle on OCI with automated model switching, ensuring 99.9% SLA compliance.
- **Scalable AI Backend:** Optimized FastAPI/Docker backend for multi-model orchestration, reducing latency by ~200ms.

CERTIFICATION

- Oracle Cloud Infrastructure 2025 Data Science Professional – Oracle | 2025
- AWS Certified AI Practitioner – Amazon Web Services | 2025

EDUCATION

M.Sc. Data Science Vellore Institute of Technology (VIT), Vellore CGPA: 8.44	2022 - 2024
BCA (Computer Applications) Vellore Institute of Technology (VIT), Vellore CGPA: 7.27	2019 - 2022